

American University of Armenia, CSE
CS121 Data Structures A
Fall 2026

Homework Assignment 5

1. Efficiently implement the `Deque` interface using an array as an underlying structure. The class should include implementation of all the `Deque` methods as well as two constructors– one with a default capacity of 32, and one with a custom capacity. Some of the mutator methods should throw an `IllegalStateException` if the underlying container is full.
2. Write a single generic method that both removes every second element and reverses the contents of a queue.

Add a main method to test it on an `LinkedList` object.

3. Implement the Queue ADT using 2 stacks as the underlying container. Enqueue should be $O(n)$, first and dequeue should be $O(1)$.